

Opinion

What biology can and cannot tell us about conscious AI

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Progress in artificial intelligence is turning machine consciousness from a philosophical curiosity into a societal issue, and has led to criticism of the widespread computational functionalist framework. Biological Naturalism (BN) claims that biology, not computation, is crucial for consciousness. In this opinion article, we highlight a dichotomy between two types of BN: Constitutive-BN and Enabling-BN. This dichotomy serves to distinguish possible BN theories. We argue that Constitutive-BN, which claims that biology is ‘intrinsically’ special, cannot be empirically supported, whereas Enabling-BN, which claims that biology affords unique information-processing abilities, can. Importantly, Enabling-BN is compatible with computational functionalism. We conclude that biology, by itself, cannot tell us which systems are conscious. However, it can act as a guide toward the required information processing.

Significance

AI progress is turning artificial consciousness into a societal issue. In this context, Biological Naturalism, the view that only biological systems can be conscious, is gaining traction. We ask which forms of Biological Naturalism can be grounded in empirical research, thereby avoiding unresolvable theoretical disputes. Empirically testable forms are not necessarily incompatible with other frameworks. Biological Naturalism can therefore remind us to take biology seriously, but it cannot currently rule out artificial consciousness.

Machine consciousness as an empirical question

Artificial intelligence (AI) is rapidly acquiring sophisticated cognitive abilities, including language, vision, memory, attention, autonomy, and even forms of embodiment. These tantalizing advances have revitalized the age-old question of whether machines can be conscious, turning what was once a theoretical curiosity into a pressing scientific and societal issue.

Within the scientific community, there is broad agreement that current AI systems are not conscious, but far less agreement on why [1,2]. Importantly, many of the limitations of current systems could, in principle, be overcome soon [2]. For example, embodiment has been raised as a potentially necessary condition for consciousness [3], a condition that the most advanced AI systems do not currently satisfy. However, there are, in principle, no barriers to this limitation being overcome in the coming decades. To bring the debate beyond necessary conditions that might soon be satisfied, it is essential to reach consensus on a clear scientific framework for determining which systems can be conscious.

Computational functionalism [4] has been the dominant framework in recent decades. On this view, the mind and consciousness can be understood as ‘software’ running on the brain’s biological ‘hardware’. Computational functionalism characterizes consciousness in terms of computational organization: consciousness arises when the ‘right’ kinds of computations occur. This framework is well in line with cognitive neuroscience, which studies neural computations within the framework of the brain as an **information-processing** system [5]. However, the advent of AI systems reproducing human cognitive abilities has shaken the foundations of computational functionalism [1] [6,7].

Against this backdrop, **Biological Naturalism (BN)**, often framed as an alternative to computational functionalism, is a perspective that has recently been gaining traction [1,6,7]. BN shifts the focus from computation to the biological substrate. Consciousness occurs only in the right kind of biological systems. While this view comes in many variants, these variants share the intuition that

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something distinctive about biology is crucial for consciousness. Hence, according to BN, current AI systems are not conscious, and prevailing AI approaches based on computer chips will never give rise to consciousness.

It is worth mentioning another recent alternative to computational functionalism based on causal structure [8], which has been criticized for being empirically hard to test [9]. This has led to polarized and potentially unsolvable theoretical clashes between paradigms [10,11] rather than evidence-based arguments. To avoid such an outcome, the debate around BN and computational functionalism must be grounded in shared standards of evidence and cumulative science. The key question is therefore not simply whether biology matters, but what empirical predictions follow if it does.

In this spirit, we adopt an empirically oriented methodological stance, asking which forms of BN are empirically testable and how. We will argue that types of BN claiming that biology is ‘intrinsically’ special cannot be empirically supported. However, types of BN claiming that biology affords unique information-processing abilities can. This clarifies what biology can, and cannot, tell us about conscious AI: it can act as a guide toward which kinds of information processing are relevant, but it cannot, by itself, determine which systems are conscious. Crucially, this focus on (biological) information processing is compatible with computational functionalism, challenging the traditional opposition between these views.

Setting the stage

There is no question that many biological features are necessary conditions for human consciousness. Nevertheless, if our goal is to answer whether machines can be conscious, we need to ask whether these features are important for consciousness in general [12]. **Multiscale integrated processing** [6] and **astroglia** [13] have been suggested as necessary for conscious processing in the brain. But it remains an open question whether these features are necessary for consciousness in general. For example, integer addition, like any cognitive function, relies on biological features in the brain [14] but can also be implemented without them in calculators.

Like any discussion about consciousness, this debate is inevitably impacted by strong intuitionsⁱ (Box 1). Can it be a coincidence that all the conscious systems we know are biological? How could predicting the next word in a language model lead to consciousness? How could machines be conscious given how different computer chips are from brains? Like all intuitions, these arguments reflect deeply ingrained cognitive heuristics, making them powerful tools for thinking but also potentially misleading [16]. As discussed in Box 1, they do not, by themselves, rule out machine consciousness. Instead, they help identify the questions that an empirical version of BN would need to answer.

A dichotomy for BN: two ways biology could matter for consciousness

There is considerable diversity among views labeled here as BN. Despite this diversity, we can distinguish two types, which we will refer to as Constitutive-BN and Enabling-BN, depending on whether biological systems are proposed to have information-processing abilities that other systems do not possess (see [1] for an alternative classification of BN theories). This dichotomy serves as a wedge between possible variants of BN and will help clarify which ones can be studied empirically. This distinction is exhaustive, in the sense that any BN proposal necessarily falls into one of these types: either biological systems are proposed to have unique information-processing abilities or they are not. Note also that it is not always clear to which type a given BN proposal belongs, as these views are constantly being developed. As such, we do not aim to classify existing BN proposals, but rather to provide a framework to help determine which types of BN are empirically testable.

Glossary

Access consciousness: information that is available for cognitive functions such as reasoning, decision-making, memory, verbal report, and the control of behavior. Unlike phenomenal consciousness, access consciousness is defined by what information can do within the cognitive system rather than by its subjective feel.

Astroglia: a class of nonneuronal brain cells that support and regulate neural activity by maintaining the chemical environment, supplying energy, and interacting with synapses.

Autopoiesis: the capacity of a living system to continuously build and maintain its own parts and its own boundary (‘self-making’). A living cell, for instance, constantly renews the very molecules and membrane that keep it alive. It has been proposed as a defining feature of life that machines lack and, by some authors, as a necessary condition for consciousness.

Biological Naturalism (BN): the view that consciousness is a biological phenomenon—it arises only in living systems from properties or mechanisms specific to them. Variants disagree on which biological features matter, and why.

Bispectral Index (BIS): a numerical measure, derived from electroencephalography, that is widely used to monitor the depth of anesthesia. Higher values indicate wakefulness, while lower values indicate progressively reduced levels of consciousness. Although calculated from brain activity, the BIS was developed and validated by calibrating its values against behavioral signs of responsiveness during anesthesia.

Chaotic system: a deterministic system that is highly sensitive to its initial conditions. Tiny differences can grow exponentially, making precise long-term prediction impossible even though the rules are both fixed and known. Weather is the classic example: its exact trajectory is, in practice, unpredictable, yet it settles onto a constrained set of states (an attractor) that keeps its long-run statistics stable—so models can still capture its important features.

Computational functionalism: the view that consciousness depends on what a system computes (or does), not on what it is physically made of. The mind can be seen as software that could, in principle, run on different

Box 1. Intuitive arguments are insufficient to establish the impossibility of artificial consciousness without empirical testing

Debates about consciousness are strongly shaped by widely shared intuitions, and several familiar arguments for denying consciousness to machines draw on them. We argue that these arguments do not, by themselves, rule out machine consciousness but can help identify empirical questions that BN must answer.

Intuition 1: induction from known biological cases

All conscious systems that we currently know of are biological. From this, it may be tempting to infer that biology is crucial. However, showing that all known conscious systems share property X does not establish that X is crucial for consciousness. For example, all conscious systems we know of live on Earth, but that is not an argument showing that only the Earth can support consciousness. Hence, which biological features (if any) are crucial for consciousness cannot be settled by induction from biology alone, but must be argued for and tested empirically.

Intuition 2: the redescription fallacy

All systems are sometimes deemed incapable of consciousness because they are ‘just’ doing something simple. For example, they are ‘just trained to predict the next word’. By the same logic, humans could be described as ‘just being selected to pass on their genes’. As Millièr and Buckner [15] put it:

‘[The redescription] fallacy arises when critics argue that a system cannot model a particular cognitive capacity ϕ , simply because its operations can be explained in less abstract and more deflationary terms. [...] Such arguments are only valid if accompanied by evidence demonstrating that a system, defined in these terms, is inherently incapable of implementing ϕ . To illustrate, consider the flawed logic in asserting [...] that brain activity could not possibly implement cognition because it can be described as a collection of neural firings’.

Hence, redescriptions are insufficient on their own to deny artificial consciousness. Arguments and empirical data are needed to show which consciousness-relevant features (if any) are impossible for artificial systems.

Intuition 3: from difference to exclusivity

There are real differences between biological and artificial systems. For example, computers are based on discrete computation, which differs from the continuous, analog nature of biology. BN proponents claim that these differences are important for consciousness. Here, one must be wary of the motte-and-bailey fallacy, which is a rhetorical move in which one advances a strong claim (the bailey) but, when challenged, retreats to a weaker claim (the motte). Thus, the strong claim is never actually defended. In the case of BN, one cannot advance a strong BN claim (only biological systems can be conscious) while, in practice, only defending a weaker claim (there are differences between artificial and biological systems). There is a huge gap between these claims, since differences between biological and nonbiological systems may not matter for consciousness. Arguments and empirical data are needed to show which differences (if any) matter.

Constitutive-BN

Constitutive-BN proposes that biology does not offer any special information-processing abilities that nonbiological systems do not have. That is, we could, in principle, replicate the brain’s information processing in machines. Nevertheless, only the biological implementation would be conscious. According to Constitutive-BN, what matters is what biology is made of, not what it does.

One version of this view is based on the idea that two systems can be equivalent computationally but differ subcomputationally [7]. For example, an AND gate is a very basic computational system with two inputs and one output: if both inputs are turned ‘on’, the output turns ‘on’, and otherwise the output is ‘off’. This information processing can be implemented equivalently in an electrical circuit (each input provides 1 V, and if the total voltage is 2 V, the output is ‘on’) or in a system using water pumps (replace electricity with water and volts with water pressure). Both implement the same information processing but differ subcomputationally. Constitutive-BN proposes that biological **subcomputational features** matter for consciousness rather than features of information processing.

We argue that this claim cannot, even in principle, be tested empirically, for reasons related to the ‘**unfolding argument**’ [9] (see alsoⁱⁱⁱ). Indeed, empirical tests of consciousness all ultimately rely

hardware. A system would be conscious when it carries out the right kind of information processing, whether this processing is implemented biologically or artificially.

Global Workspace Theory: a scientific theory of consciousness that proposes that information becomes conscious when it is globally broadcast across a widespread network of brain regions through a process called ignition. In contrast, unconscious information remains confined to more local processing.

Information processing: the transformation and use of signals within a system—for example, to integrate sensory inputs, update internal states, guide behavior, or produce reports. In this opinion article, the key question is whether biological mechanisms matter for consciousness because they enable specific forms of information processing.

Multiscale integrated processing: the idea that biological brains perform computations through tightly coupled interactions across multiple spatial and temporal scales—from molecules and synapses to neurons, circuits, and whole-brain dynamics. This multiscale integration has recently been proposed as a distinctive feature of biological computation that current digital computers lack and as a potential requirement for consciousness.

Overflowing phenomenal consciousness: the view that phenomenal consciousness contains more information than can be accessed for report, memory, or cognitive control. According to this proposal, people may consciously experience rich details that are nevertheless unavailable for report or memory. This idea remains the subject of substantial debate in consciousness science.

Phenomenal consciousness: the subjective, first-person aspect of experience—what it feels like to see a red apple, hear a melody, or feel pain. It refers to the qualitative character of experience itself (‘what it is like’), independently of how that experience is used by the cognitive system.

Subcomputational features: physical properties of a system that lie beneath its computations. A mechanical clock and a digital watch keep the same time by entirely different physical means; the difference between them is subcomputational.

Unfolding argument: first raised against Integrated Information Theory,

on behavioral reports^{ii,iii} [17–22]. When studying the contents of consciousness, we present stimuli and ask observers to report what they perceive. To study conscious states, we rely on behavioral measures too (e.g., by asking a patient to respond to questions). Even physiological measures, such as the **Bispectral Index (BIS)** used to assess depth of anesthesia, need to be calibrated against behavioral measures. Yet Constitutive-BN grants that biological and nonbiological systems could, in principle, implement the same information processing, leading to identical behavior, and therefore identical experimental results about consciousness, while still maintaining that only biological systems will be conscious.

To illustrate the problem, imagine a scenario in which part of a patient's brain that is crucial for consciousness is replaced by a digital computer chip. If the chip performs equivalent information processing (which is possible by hypothesis in Constitutive-BN), the brain as a whole operates as usual, leading to identical reports about consciousness. Yet Constitutive-BN claims that consciousness has disappeared or been altered. The problem is that this claim cannot be supported by experimental evidence, since nothing observable has changed in the subject's reports about their consciousness. The patient still insists on being conscious and behaves identically in all experiments. But Constitutive-BN claims that none of this matters because biology is absent—an empirically untestable claim (see [9,18,22],^{iv} [23,24] for related discussions).

As far-fetched as this example sounds, we are getting closer to real-world cases. Researchers recently implanted electrodes in the occipital cortex of a blind patient and successfully evoked simple conscious visual percepts [25] (more detailed percepts were evoked in monkeys [26]). How do we know that conscious percepts were evoked? We know this because the patient reported experiencing them. But Constitutive-BN implies the possibility that because the relevant biological aspects might not have been in place, the patient might merely have reported these percepts without experiencing them consciously. The problem is that, under Constitutive-BN, we can never know whether this is the case. The Constitutive-BN proponent must always doubt such reports on the grounds of the nonbiological nature of the implant, no matter how reliably the patient claims to be conscious.

While this is still a rudimentary case, more powerful implants are increasingly plausible, and the field of brain-machine interfaces is making great strides [27]. In the not-too-distant future, it may be possible to offload more important aspects of brain information processing to artificial systems. This could have serious impacts on consciousness that, according to Constitutive-BN, can never be detected empirically, even in principle.

In summary, Constitutive-BN implies that consciousness cannot be detected through empirical measurements, but should be inferred from the biological substrate. In principle, this could be true, but we can never gather any scientific evidence to support it, making it scientifically inert (see also [18,21] for related arguments, and the debate around **overflowing phenomenal consciousness** [28]). This precludes empirical guardrails, likely leading to unresolvable fragmentation, endless debates, and ultimately a loss of credibility for the field [29,30].

Enabling-BN

Enabling-BN proposes that biology offers unique information-processing abilities. That is, we could not implement what the brain does in nonbiological systems, even in principle, because there are key differences between biological and nonbiological systems [14]. According to Enabling-BN, what matters is what biology does, not what it is made of. For this reason, Enabling-BN proposes that only biological systems can be conscious. Two questions arise: (i) are these proposed differences between biological and nonbiological systems real? (ii) Are there reasons to believe that these differences matter for consciousness? Here, we set question

holds that two systems can always be built to behave identically in every situation yet be wired very differently internally. If a theory makes consciousness depend on that internal wiring rather than on what the system does, it must judge these twins differently—a claim that no behavioral test could check. The paper applies analogous reasoning to some variants of BN.

(i) aside and accept, for the sake of argument, that biological systems have unique abilities in order to focus on the importance of those abilities for consciousness.

To make this point more concrete, we consider one frequently cited difference between biological and artificial systems: digital computers operate on discrete bits of information, while biological systems exhibit continuous analog dynamics over time (spike timing is not synchronized to a discrete clock cycle, unlike in digital computers), space (membrane potentials and chemical gradients vary continuously), and magnitude (membrane potentials can take any continuous value) [1,7]. This difference may matter in **chaotic systems**, where even infinitesimally small differences in initial conditions can lead to large divergences over time [31]. Hence, the continuous nature of biological dynamics, compared with the discrete nature of digital computers, has been proposed as a relevant difference by proponents of BN: because digital systems represent states with finite precision, they cannot reproduce exactly the trajectory of a continuous chaotic system, and small discrepancies may lead to widely different outcomes [1].

Crucially, however, the existence of a difference between biological systems and computers does not entail that this difference matters for consciousness. Chaotic systems do not generate arbitrary behavior. Their dynamics are constrained by attractors that determine the trajectories the system can follow. Digital systems can therefore diverge from their continuous counterparts while still preserving the long-time statistical behavior. Climate science provides a useful analogy. Weather systems are chaotic, meaning that even tiny numerical differences quickly lead to vastly different weather predictions. Hence, we all agree that the analog nature of the atmosphere is different from the discrete nature of computers. However, despite these differences, computers can capture important features of weather systems, and this is precisely what makes weather and climate modeling scientifically useful. There is a separate debate on whether a simulated hurricane is ‘wet’ [1].^v However, this is beyond the scope of this opinion article; what matters for our argument is that discrete computers can capture at least some important features of hurricanes, such as their size, speed, formation dynamics, or the eye of the storm^{ii,v,vi} [32]. Likewise, despite real differences between the analog nature of the brain and the discrete nature of computers, the latter may still be able to capture what matters for consciousness. Alternatively, they might not be able to capture some crucial features of consciousness, but arguments and experiments are needed to support this claim (see the next section).

As another example, **autopoiesis** has been argued to be a defining property of living beings that is absent in current AI systems [1]. Assuming that autopoiesis is indeed uniquely biological and enables specific information-processing capacities, such as embodied regulation or organism-level integration, it constitutes a valid difference between biological and nonbiological systems. But the existence of this difference does not show that it matters for consciousness. To become empirically testable, the proposal must specify which capacities autopoiesis enables, and how these capacities should affect conscious processing. Alternatively, if autopoiesis is treated as sufficient for consciousness independently of any observable consequences, the proposal collapses into Constitutive-BN and faces the corresponding challenges.

This argument generalizes to any other proposed difference between biological and nonbiological systems. Even accepting that there are differences, it is not enough to just point out that they exist. We need to empirically show that they matter for consciousness.

Importantly, an Enabling-BN proposal needs to answer both questions (i) and (ii) above: the feature proposed to be crucial for consciousness needs to be (i) unique to biology and (ii) relevant to consciousness. Several recent proposals take steps in this direction. For example, dendritic

integration has been experimentally linked to consciousness [33] (addressing question ii), but it has not been shown that the features of dendritic integration crucial for consciousness cannot be implemented in nonbiological systems (leaving question i open). Conversely, there are arguments proposing that certain aspects of biology may be extremely difficult to implement in machines [6,14] (addressing question i), but it has not been shown that these same uniquely biological features matter for consciousness (leaving question ii open).

In summary, we argue that Constitutive-BN does not generate empirically testable predictions about consciousness. In contrast, Enabling-BN can be safely grounded in empirical research, provided that proponents specify which differences between biological and nonbiological systems matter for consciousness and test these claims empirically. In the absence of empirical evidence directly testing the involvement of uniquely biological features in consciousness, BN remains a theoretical possibility but is not a supported theory.

What evidence links biology to consciousness?

What would empirical tests of BN look like? In consciousness science, empirical evidence comes from contrasting conscious versus nonconscious processing [17,22].^{vii} This applies to both the state (e.g., dreamless sleep vs. dreaming) and content (e.g., seen vs. unseen targets in psychophysical experiments) of consciousness. We can gather these empirical data even in the absence of a full consensus on how exactly to define consciousness theoretically. Science can start with good observations and aim for definitions [17]. Hence, for the moment, we set aside theoretical debates that do not impact empirical data, for example, about the relationship between **phenomenal consciousness** and **access consciousness** (see [Glossary](#)) [28]. Whatever the final answer, a claim that biology is necessary becomes empirically tractable only if the relevant biological property impacts conscious processing in ways that can be experimentally assessed.

Theorists need to show that when a variable proposed to matter for consciousness changes, consciousness changes as predicted. For example, **Global Workspace Theory** [34] can predict [35] that a briefly presented target will remain unconsciously represented when activity stays localized to sensory areas but will be consciously perceived when it triggers ‘ignition’: a sudden, self-sustaining spread of activity across a wider brain network [34–36]. While there remain important debates about such experiments [37], they illustrate how a theory can gain empirical traction by linking a proposed mechanism to specific contrasts between conscious and nonconscious processing.

As discussed, several theoretical candidate differences between biological and nonbiological systems have been suggested, ranging from dendritic integration [38] to multiscale brain dynamics [6]. To gather empirical support, BN needs to pinpoint aspects unique to biology and show that they are crucial for consciousness. It should predict: ‘when this biological feature X is absent, there is no consciousness; when it is present, there is consciousness’. Returning to the example of continuous versus discrete computations discussed in the previous section, BN needs to explain why this matters. Is it because of chaos? If so, why is chaos crucial for consciousness, given that it is not crucial, for example, for the formation of hurricanes? One possibility would be to predict that, when the system is at the wrong location on its chaotic attractor, there is no consciousness; when it is at the right location, there is consciousness. This prediction then needs to be tested. Analogous reasoning applies to other proposed relevant differences between biological and nonbiological systems.

Biology plays an enabling role, not a fundamental one, in testable BN theories

Constitutive-BN and Enabling-BN take fundamentally different views on how to determine whether only biological systems can be conscious. Constitutive-BN proposes inferring

consciousness from the biological substrate, whereas Enabling-BN asks which biological properties affect information processing relevant to consciousness.

The above considerations lead to two conclusions: (i) BN can be safely grounded in empirical science, but (ii) only if the relevant biological properties have an impact on information processing. Hence, BN cannot support the claim that biology is intrinsically special. What matters is what the biology does (Enabling-BN), not what it is made of (Constitutive-BN). What matters is not the substrate *qua* substrate, but that it may enable information processing abilities that are impossible or inefficient in current computers.

This reframes the traditional opposition between BN and computational functionalism. Arguments against computational functionalism are often taken to count as arguments in favor of BN [1,7]. However, Enabling-BN and computational functionalism are not mutually exclusive [6]. Both could be true if the forms of information processing required for consciousness can only be implemented by biological systems. Both could also be false: consciousness might depend on properties that are not captured by abstract computation but are not exclusive to living systems either. The two views can also come apart: computational functionalism could be true while BN is false if the relevant computations can be implemented nonbiologically, whereas BN could be true while computational functionalism is false if consciousness also depends on biological properties beyond information processing (i.e., Constitutive-BN, but this is not empirically testable for the reasons discussed earlier). Hence, arguments against computational functionalism should not be taken to support BN, and vice versa.

The argument can be summarized as follows: the biological implementation of human consciousness either contributes to its information processing (Enabling-BN) or it does not (Constitutive-BN). If it does contribute, then BN is testable via that contribution and is compatible with computational functionalism. If it does not contribute, then BN is incompatible with computational functionalism and is untestable. The conclusion is that BN is either compatible with computational functionalism or, if incompatible, is untestable.

Many contemporary BN proposals adopt an Enabling-BN view by treating biological features as enabling specific kinds of processing crucial for consciousness rather than as intrinsically consciousness-making substances [6,14]. Our aim is not to reject these proposals. Instead, our contribution is threefold: (i) we propose that the Constitutive/Enabling distinction is essential to understanding which types of BN are testable, (ii) we provide an argument against the empirical testability of Constitutive-BN, and (iii) we explicitly state a methodological requirement for Enabling-BN: such proposals become scientific hypotheses only when they specify which biological capacities matter, how they affect conscious processing, and how these claims could be tested.

This opinion article asks what biology can, and cannot, tell us about conscious AI. We argue that biology, by itself, will never be able to determine whether a system is conscious. Therefore, we should not base our decisions about future AI on whether they are biological. The central challenge is still to understand which kinds of information processing matter for consciousness. Biology can act as a guide in this quest, but not as an answer.

Concluding remarks

The rapid rise of AI raises the question of artificial consciousness and poses huge ethical challenges that we should address before it is too late. A common argument for BN is that we

Outstanding questions

Which biological features, if any, are necessary for consciousness beyond their roles as incidental realizers of mammalian cognitive function?

What experimental paradigms could distinguish 'biological feature X is necessary for consciousness' from 'biological feature X is sufficient in the mammalian implementation as we know'?

Could nonbiological substrates (neuromorphic implementations, brain organoids, and synthetic biology systems) serve as test beds for substrate-dependence claims?

If consciousness depends on biologically specific information processing, does this define a subclass of computational functionalism or a category genuinely outside it?

How should the risks of wrongly attributing consciousness to AI (false positive) and wrongly denying it to nonbiological systems (false negative) be weighed in policy and ethics?

should avoid false positives: wrongly attributing consciousness to machines [1]. This concern is legitimate, but attributing consciousness too readily to biological systems such as embryos raises equally challenging ethical issues. Hence, consciousness should not be given to machines ‘for free’, but it should not be taken as a basic biological property without specifying the relevant mechanisms either.

To make progress on these questions, we need to understand which differences between brains and current AI are crucial for consciousness. This is a deep challenge, as current AI systems differ from humans in endless ways. Biological implementation is indeed one of them, but from a computational perspective there are enormous differences too. Hence, there are other options besides BN to demonstrate differences between biological and AI systems. BN should be held to the same high standards of empirical testing as computational proposals. Currently, there is arguably little empirical evidence for a crucial role of uniquely biological features in consciousness. Deriving paradigms to test this should be a priority for BN. Community-wide events and adversarial collaborations [39] are promising approaches to do so.

Importantly, empirically testable Enabling-BN is not incompatible with computational functionalism. Hence, we should not spiral into an entrenched clash of paradigms based on theories and intuitions. BN draws our attention to biological processes, while computational functionalism focuses on the computer metaphor of the brain. But these views do not offer any easy ultimate solutions to the problem of consciousness. We all still face the same daunting task: understanding which kinds of information processing matter for consciousness.

We are living through strange and puzzling times, and we should be open to new ideas. But crucially, consciousness and AI researchers can solve these questions better if they work together (see [Outstanding questions](#)). We should not isolate consciousness science from the rest of science. We should not entangle ourselves in a veil of biological exceptionalism. As the eyes of the world turn to the science of consciousness for answers, we need to avoid dogmatism and try to reach a consensus. To this end, it is crucial to ground debates in empirical evidence.

Author contributions

Conceptualization: U.K. and A.D.; methodology: U.K. and A.D.; writing—original draft: U.K. and A.D.; and writing—review and editing: U.K. and A.D. Both the authors contributed equally to all aspects of the manuscript.

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Declaration of interests

The authors declare no competing interests.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Claude to correct spelling and grammar and to improve readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Resources

[†]<https://philpapers.org/rec/LERTAF>

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Seth argues that consciousness depends on properties of living organisms and is therefore unlikely to emerge along current AI trajectories, though brain-like or life-like artificial systems may be more plausible candidates. His article provides a prominent contemporary defence of Biological Naturalism.

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Block argues that consciousness might depend not only on computational roles but also on the subcomputational biological mechanisms that realize them. This can be interpreted as an influential example of constitutive-BN.

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Milinkovic and Aru argue that biological computation differs from digital computation through scale-inseparable multiscale processing and hybrid discrete–continuous dynamics. It is an influential example of Enabling-BN, proposing a concrete biological difference while leaving open whether that difference matters for consciousness.

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Butlin and colleagues derive indicators of consciousness from neuroscientific theories and use them to grade particular AI systems. This is an influential example applying the computational functionalism framework to answer questions about AI consciousness.

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Author biographies

Ulysse Klatzmann is a postdoctoral researcher at Université de Montréal and CHU Sainte-Justine. His work uses large-scale dynamical and spatial modeling to study how cortical organization supports cognition and the conscious perception of information. More broadly, he is interested in how computational neuroscience can help clarify theoretical debates in consciousness science, including questions about artificial consciousness. He is also a founding member of the Mediterranean Society for Consciousness Science (MESEC), an international early-career network that promotes interdisciplinary collaboration among researchers in training in consciousness science.

Adrien Doerig is a professor at Freie Universität Berlin, where he leads the Cognitive Computational Neuroscience Lab. His research combines neuroscience, machine learning, psychophysics, and philosophy to study perception, visual cognition, and consciousness. He is especially interested in how artificial neural networks and computational models can be used to formulate and test theories of brain function. His work also examines the empirical status of theories of consciousness, including how theoretical claims can be linked to measurable predictions.

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Caviola and colleagues use attitudes to animal consciousness to forecast how society will respond to AI consciousness, expecting disagreement rather than consensus, driven partly by substratism: favouring carbon-based beings over silicon-based ones. Such intuitions are what an empirically grounded criterion would have to replace.

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